The O(1)-State Organism: Seven Measured Foundations for Continuously Living Sequence Models

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Every number in this paper is committed in the public repository ***github.com/DT-Foss/o1-state*** *with its generating script and raw JSON; the prediction ledger (****analysis/PREDICTIONS.md****) was written before the data it scores. This draft is itself part of the public disclosure.*

Abstract

Contemporary sequence models are batch artifacts: trained on a frozen corpus, frozen at deployment, and structurally incapable of one thing no scaling law addresses — **they never experience their own future**. We present measured foundations for the alternative: a bounded-state organism that consumes an unbounded stream at constant memory, decides *from its own prediction error* when to learn, what to remember, when to consult an external index, and when to sleep — and whose complete living state is a small, portable, forkable asset. In a pre-registered 40-hour experiment (909.7M streamed tokens, 16/16 integrity checks), the organism's surprise gate did not merely approach full-gradient training at 25.2% of the gradient tokens — it **beat** it (improvement ratio 1.0091), crossing the prediction's own registered "embarrassment threshold." A twin forked mid-run showed that restarting a living stream is *free* (surprise excess 0.0029, convergence one chunk after the fork) — one of three independent measurements establishing a **two-timescale law**: the fast state rebuilds within a receptive field and carries no durable value, while everything worth keeping lives in slow carrier channels, weights, and an external index. We further measure: exact detach-carry streaming (gradient cosine 1.0), keyed holographic recall through silence with a movable persistence knee ($32 \rightarrow 2176+$ via an interaction of two eval-time levers), dosed sleep consolidation with a measured life curve, collective memory (one organism's stored surprises cut another's shock-forgetting to $0.67\times$), live cross-architecture migration with behaviorally identical continuation to six decimals, and family-generic transfer of the entire operating mode to a Mamba/S6 parametrization ($0.98\times$). Thirty-nine predictions were registered before their data; the falsifications are reported at the same strength as the confirmations. We argue these primitives compose into a **reality prediction engine**: a system that lives on streams of the present, deposits predictions about its own future at every horizon, and learns from their arrival — an axis of capability orthogonal to parameter scaling, and one on which batch models cannot compete by construction.

1 The blind spot

A language model trained on a fixed corpus optimizes next-token prediction on the *past*. At deployment its weights are dead; its context window is a purchased illusion of memory; a session's experience evaporates. Three structural consequences follow. (i) The model cannot *verify* its own predictions — verification requires living until the predicted moment arrives. (ii) It cannot *accumulate* — its state is either bounded-and-overwritten or unbounded-and-unaffordable. (iii) It cannot *move* — the serving process is welded to the hardware that holds its KV cache.

This paper measures the alternative at small scale and full methodological rigor. The claim is not that a 128-dimensional organism rivals a frontier model at language; it does not. The claim is that the **operating layer** every persistent deployed model will need — when to learn, what to keep, how to move, how to sleep, how to verify — is measurable *now*, that its laws are architecture-generic, and that they are already surprising: the learner's own prediction error turns out to be not merely a sufficient control signal but a *better* teacher than learning from everything.

2 The substrate (prior work, compressed)

The organism runs on GSSM-Selective, a gated bounded recurrence $z_t = \gamma_t z_{t-1} + a_t$ in which Mamba/S6 [3], S5 [7], and LRU [5] are switch restrictions of one affine prefix-scan operator (family reduction measured at ∼1e-15; repository §1). Three prior structural results carry the present work: **(a)** the operator is shift-equivariant with a ∼5–8-token contraction receptive field, giving flat perplexity at 4096× the training length and a billion streamed tokens at 4.36 GB flat RSS; **(b)** truncated-BPTT with a detached carried state reproduces full-window gradients exactly (cosine 1.0000, max-abs-delta 0.0) — streaming *training* is exact, not approximate; **(c)** gated readouts have a sharp capacity cliff at load ≈ 1 (slope 1.32 vs. 0.57 linear), which *forces* the two-system split between a bounded live state and a growable external index.

3 Seven foundations

Each is stated over the operator class and anchored by committed measurements. (Full formal statements: `FOUNDATIONS.md`.)

F1 — The surprise calculus. The learner's own prediction error suffices as the universal control signal: when to learn (gate), what to remember (spike-selected spans), when to consult (retrieval triggers), when and how much to sleep (dividend-monitored replay), how curious to be. *Anchor:* the 40h result of §4. *Extension (disclosed, first experiment registered):* a ladder of horizons — deposit predictions about the stream's own future, score them on arrival, learn at every timescale.

F2 — The exactness license. Bounded contraction makes detach-carry streaming exact and decouples training layout from deployment layout. Twelve independent equivalence measurements across every deployment primitive (train, grow, graft, swap, regularize, re-read) all pass at 0.0–1e-6.

F3 — Phase–magnitude separation. In complex bound states, content (phase) is written and never evolves (zero-drive invariance $|\Delta\varphi| \approx$ 1e-8, machine precision); persistence (magnitude) decays. Real fillers are *double agents* — they pollute the phase (removable exactly by an eval-time clamp: drift 0.0 rad) while *feeding* the magnitude (clamped, channels starve). The recall knee moved 32 → 256 → 512 → 2176 mean / 4096 end-of-range across theory-led interventions, the last requiring BOTH levers jointly — an interaction, not a sum. Knee *position* is seed-dependent; the intervention *effects* replicate.

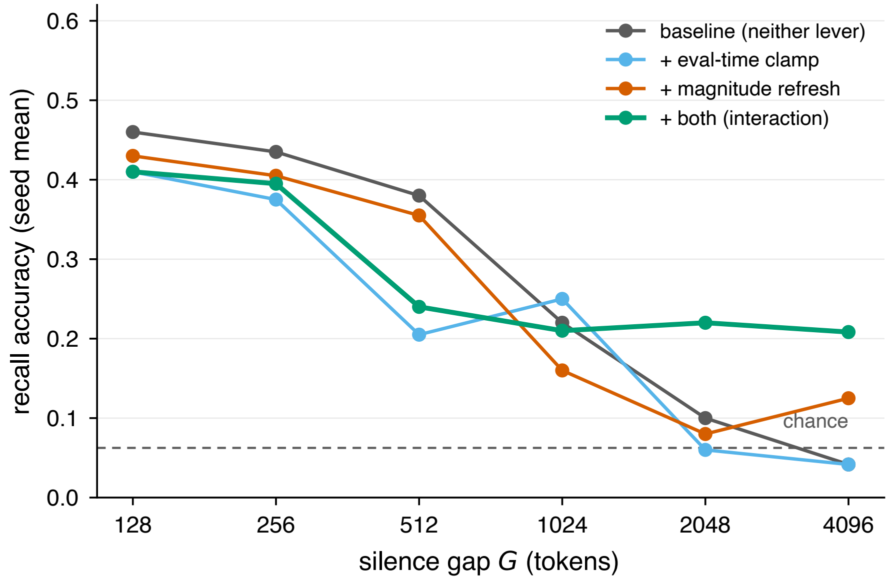

Figure 1: The persistence knee moves with the eval-time levers. Each lever alone decays toward chance beyond $G{\approx}512$–1024; applying both together holds recall at more than 3× chance out to $G{=}4096$ — an interaction, not a sum. Source: `results/holo_clamp_refresh.json`.

F4 — The two-system law. The gated-readout cliff (rank-1-per-channel binding, $D_{\text{eff}} \approx 1.02$ measured against a closed-form bound) forces unbounded accumulation into an external index. Runtime consultation lifts recall far past state capacity (0.10–0.20 → 0.41–0.62 at 16 pairs, dose-dependent, random-injection controlled); models trained *with* consultation read reminders at 0.99–1.00.

F5 — Family-generic operating modes. The full POS recipe run on a scan-parameter-matched Mamba/S6 configuration transfers at 0.98× (three seeds, two CPU architectures) — and the house operator beats S6 head-to-head (+0.13 to +0.16 nats) at parity. The calculus attaches to the operator class, not to one network.

F6 — Train short, deploy unbounded. One shift-equivariance symmetry on two axes: sequence length (flat to 4096×; corpus-as- iterator to 10^9 tokens) and silence (keyed bindings held through gaps 128× the training horizon). Its honest boundary is measured twice: multi-chunk training is structurally gradient-blind to the write — the cure is to train inside the boundary and let exactness carry the skill out.

F7 — The portable organism. The complete living system — weights, optimizer moments, carried state, gating windows, span store, stream position, RNG — is one atomic ∼53 MB artifact, and the organs couple through *kilobytes* (spans, reminders, deltas), not activations. Measured: local checkpoint→new-process→resume is bit-identical; **live mid-stream migration from Apple Silicon to x86 continues behaviorally identically to six decimals** (heldout 6.182391 == 6.182391, every gate decision matched — only the BLAS bit-digest differs, and it does not propagate); a killed replica rejoining through a *shared* span store ends *better* than its never-killed twin (−0.014: the partner's outage-window spans are a gift); an outage spent sleeping beats one spent idle (+0.065);

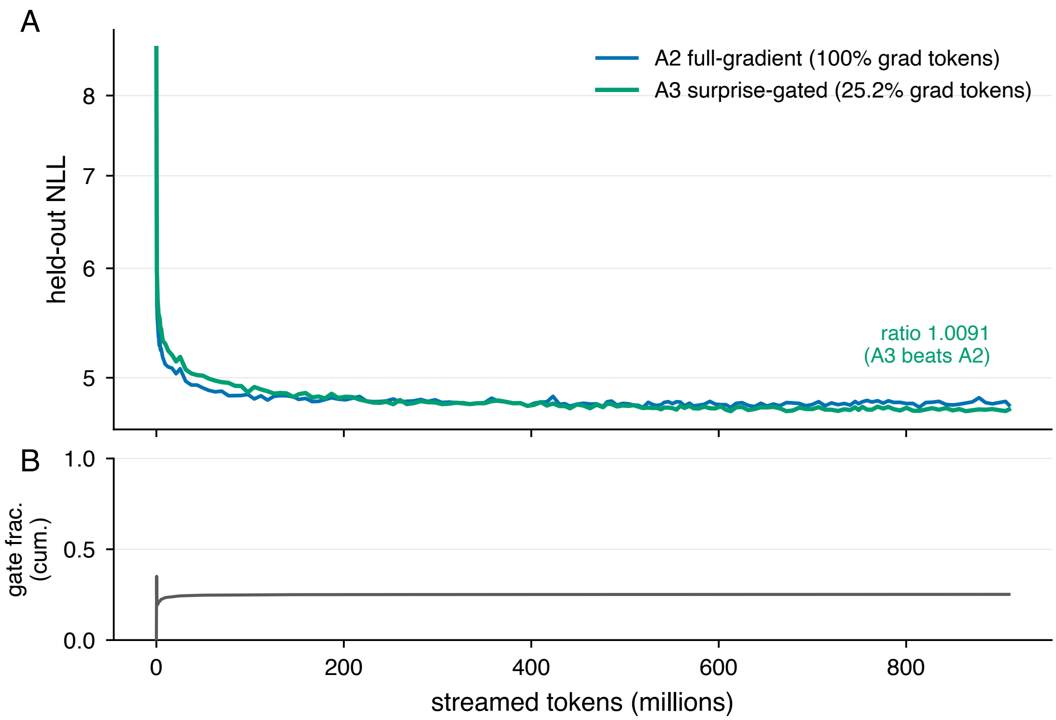

Figure 2: The 40-hour run: held-out loss for A2 (full-gradient) and A3 (surprise-gated), and A3's cumulative gate fraction, vs. streamed tokens. A3 tracks A2 while training on 25.2% of the gradient tokens and finishes ahead (ratio 1.0091). Source: `results/pos_metrics.jsonl`.

stop-free migration (source never pauses, target catches up by deterministic replay) achieves bit-identical parity at every scale tried.

4 The 40-hour experiment: the gate beats the firehose

One process, three from-scratch arms on one cloned C4 stream (seed 42), 40.0 hours, 909,676,544 tokens, no restarts, RSS $\leq$ 1.094 GB. A1: forward only. A2: backward on every chunk. A3: backward only when the chunk's own mean NLL clears the rolling 75th percentile of its own history — no oracle, no second model, no labels, no schedule.

Table 1: The 40-hour surprise-gated run vs. full-gradient training.

arm	held-out 8.6588 $\rightarrow$	improvement	gradient tokens
A2 full-gradient	4.7782	3.8806	100%
A3 surprise-gated	**4.7430**	**3.9158**	**25.17%**

Ratio 1.0091. The pre-registered prediction (P1) put the point estimate at 0.85 with an explicit embarrassment threshold: *"ratio > 1.0 sustained would be a bigger result than the thesis itself."* It fired. The surprising part is not efficiency — it is that the 75% of chunks the gate skips are, in net, *worth skipping*: selective plasticity beats indiscriminate plasticity at equal data exposure. Every number re-derives from the raw 1.78M-chunk log (16/16 integrity checks).

The twin. At T+24h, A3's weights were copied into a fresh process with zero carried state.

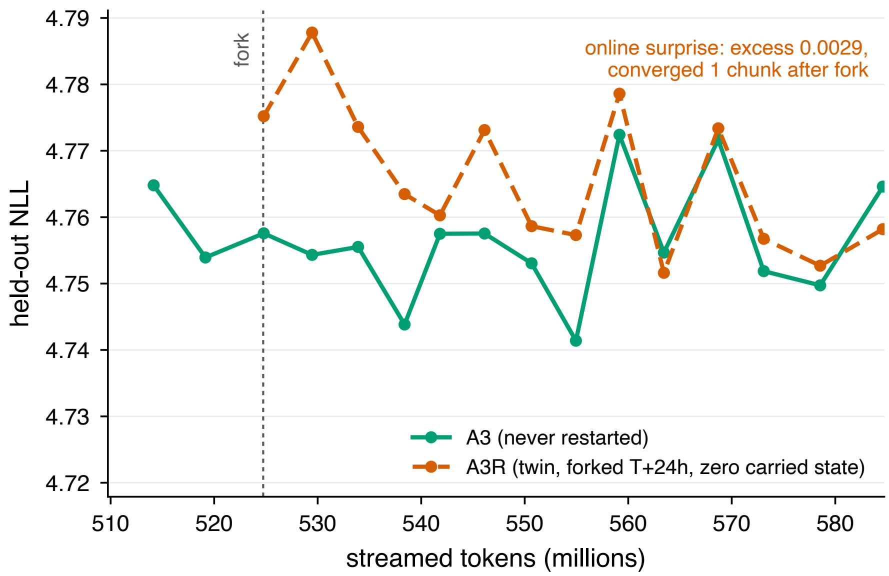

Figure 3: The twin, zoomed to the fork. A3R (forked at T+24h with zero carried state) opens ≈0.02 nats above A3, crosses into A3's own fluctuation band within a few evaluations, and finishes ahead (4.7365 vs. 4.7430). On the registered online-surprise criterion the restart is free: excess 0.0029, gate fractions identical to four decimals, convergence one chunk after the fork. Source: `results/pos_metrics.jsonl`, `results/pos_summary.json`.

Registered: a warmup tax (+0.03–0.15 excess surprise, $\geq 1.5\times$ over-gating, hours to converge). Measured: excess 0.0029, gate rates identical to four decimals, convergence one chunk after the fork, and the twin finished *ahead* (4.7365). Restarting a living stream is free at the fast timescale — the third independent measurement of the two-timescale law (§5).

The honest negative. 1020 paired index-injection probes: both deltas negative (the index froze at its 20k cap by token 11M; replaying 5–11M-era spans into an 80×-older model disturbs), but injection hurts 3.2× *less* than random injection (−0.087 vs. −0.279; helped 0.35 vs. 0.10) — the mechanism carries information; the provenance was stale. Registered as falsified; the v2 (rolling spike threshold) is specified.

5 The two-timescale law of operation

Three independent experiments give the same law. (1) The twin (§4): zero carried state costs nothing. (2) A snapshot cross-matrix (weights at 359M tokens × states from 128M/240M/359M/zero/shuffled): every arm converges to the native trajectory within 4 chunks — 256 tokens, the receptive-field scale; a cold start is not worse than the carried state. (3) The beacon: a *stored bit* in a write-once-freeze carrier channel ($\gamma \approx 0.9995$) survives a d64→d128 widening surgery at recall 1.000 and survives a weight swap across 2× training distance at recall 1.000 through 512 tokens of silence — by redundant encoding (the carrier's channel address can relocate; the read still lands).

The law: the fast path forgets by design — so weight updates, model growth (function-preserving widening: surgery equivalence 6.7e-6, no transient, growth beats restart by +0.13 to +0.17 nats, 3× replicated), and migration are *safe live operations*; everything durable lives in slow carriers, weights, and the index — exactly the layer the organism's own calculus (F1, F4) manages. Operationally this is the answer to a question every persistent-model deployment will face: *what must be preserved*

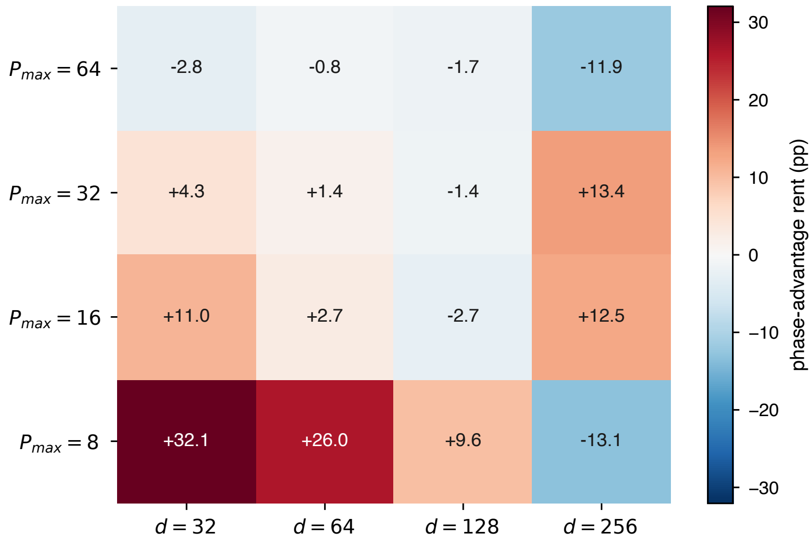

Figure 4: The 16-cell rent map: phase-advantage rent (percentage points) as a function of the recall load $P_{\max}$ and width d_{model}. The one-parameter rent law that motivated a single scalar prediction is falsified by this map's structure; the map explains, cell by cell, where the phase advantage on real text holds and where it does not. Source: `results/holo_rent_map.json`.

across an update? Measured: almost nothing — if the architecture separates its timescales.

6 The organism is social and collective

Organism A streams a C4+code mix, storing its surprise spans. Organism B, facing a code shock, replays A's shared spans and forgets **0.67**× of what it forgets with only its own spans — at better plasticity — while token-shuffled A-spans are as useless as no replay: the benefit is content, not regularization. Combined with F7: a fleet of small organisms sharing an index is a *measured* design, not an aspiration — including the striking corollary that a replica which dies and rejoins through the shared store can end *better* than one that never died.

7 Method as contribution: the adversarial ledger

Every experiment in this program was preceded by a numbered, committed prediction (P1–P39 at this writing); a scoring script audits result JSONs against the register; falsified predictions remain in the ledger with the numbers that killed them. Of the scored predictions, roughly one third were falsified or partial — including the program's own favorite hypotheses (the dream generator; rate-homeostatic curiosity; the α-regularizer; a one-parameter rent law; the phase advantage on real text — falsified *and then explained* by a 16-cell rent map measured the same night, Figure 4).

Two method-level findings emerged from the discipline itself: **training-dynamics sensitivity to floating-point reduction order near ignition boundaries** (thread count changes *whether* a model ignites, not just speed — 6/6 seeds ignite in one regime, 0/4 in another), and the resulting rule that instruments must reproduce their reference regime. We offer the ledger pattern itself as a transferable practice: a vision paper whose every claim is either a committed measurement or an explicitly open registered question.

8 The reality prediction engine (the road this opens)

The foundations compose into something none of them states alone. A system that (i) lives indefinitely on present streams at constant memory (F2, F6), (ii) controls all plasticity from its own prediction error (F1) — in the spirit of temporal-difference learning's use of prediction error as a teaching signal [8] and predictive coding's use of prediction error as the currency of cortical inference [6], (iii) can *deposit* a prediction about its own future in a persistence mechanism built for exactly that (F3's carrier; measured to survive silence, surgery, and weight updates) — the world-model commitment articulated for autonomous machine intelligence [4], (iv) scores the deposit when the future arrives — a loop only a *living* system can close, because a batch model never experiences its own future — and (v) exists as a portable, forkable, shardable, collectively- remembering asset (F7, §6): this is a **reality prediction engine**. Its scaling axis is not parameters but *scope* — more streams, more horizons, more organisms sharing more index. The first horizon-ladder experiment (early-warning under distribution shift; the long-horizon head as gating teacher) is registered (P37) and running at this draft's commit.

9 Limitations, stated plainly

Scale: $d_{\text{model}} \leq 128$, WT-2-derived vocabularies; no claim of frontier language quality is made or implied. The phase mechanism's ignition is fragile (seed- and reduction-order-dependent) — a measured fact that bounds current practice and is itself a finding. The injection result is provenance-limited (stale index era). Wall-clock throughput numbers are never claimed (host contamination); all axes are token-based. The rank-1 anchor's generating script is lost (artifact-backed, reconstruction leads documented) — reported as reproduction debt.

10 Invitation

Everything here reruns from the repository: the harnesses, the exact configurations, the raw JSONs, the ledger, and the scoring script. The organism that produced §4 can be resumed, forked, migrated, or extended by anyone — that is what F7 is for.

Related work (to be expanded in v0.2): predictive coding and JEPA (the vision lineage) [6, 4], TD-learning (multi-horizon error) [8], zoology/MQAR (recall instruments) [1], Mamba/S5/LRU (the operator family) [3, 7, 5], VM live migration (the systems analogy) [2].

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